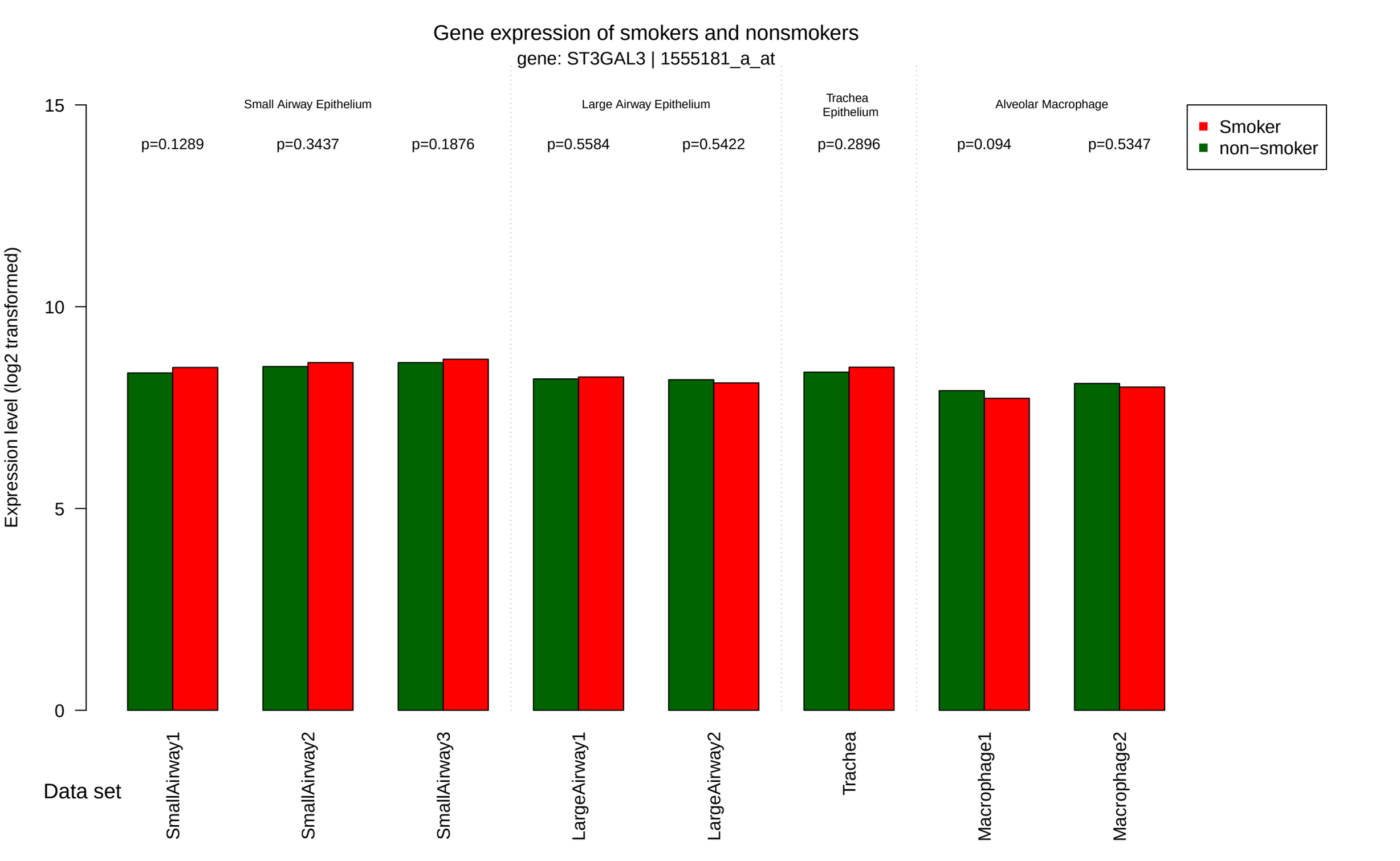
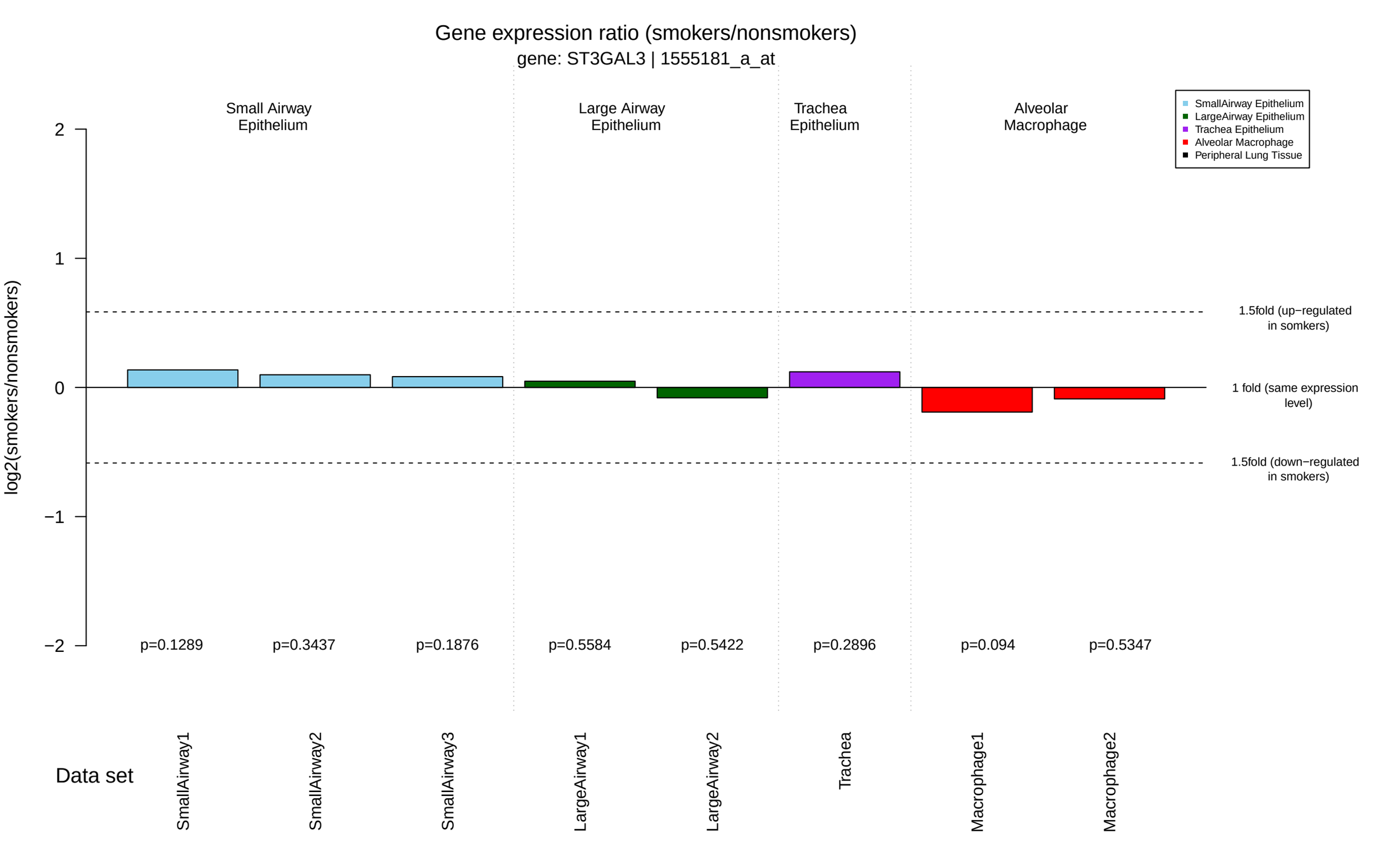
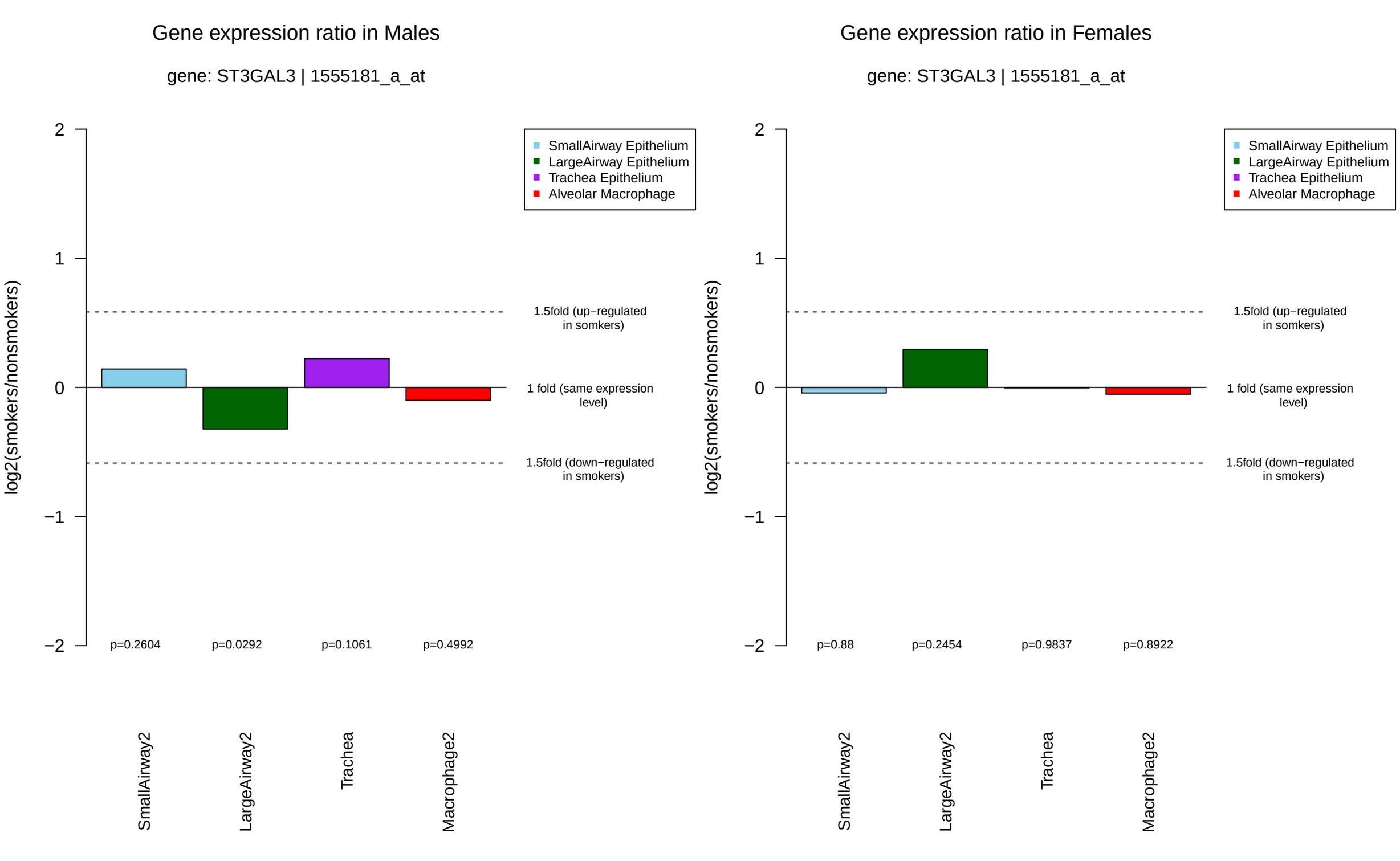
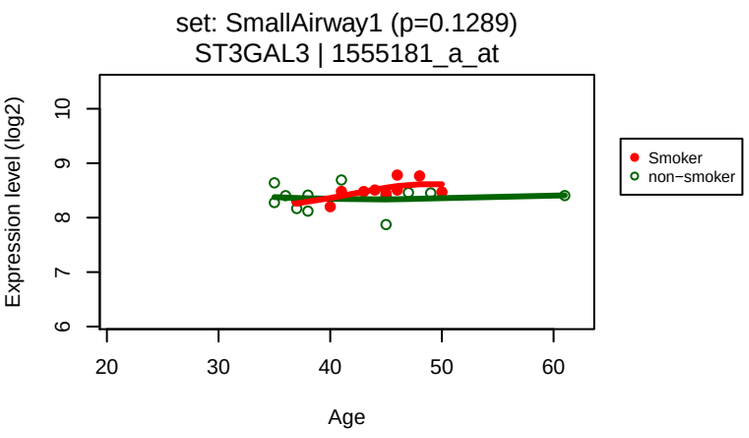
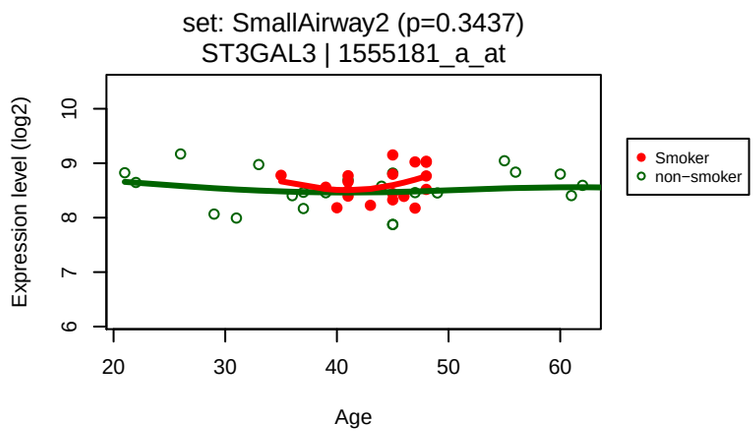
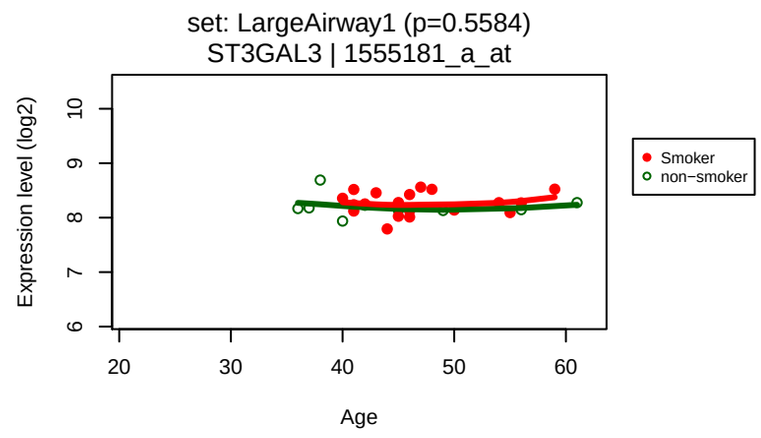

Gene expression of smokers and nonsmokers
gene: ST3GAL3 | 1555181_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1289
p=0.3437
p=0.1876
p=0.5584
p=0.5422
p=0.2896
p=0.094
p=0.5347
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

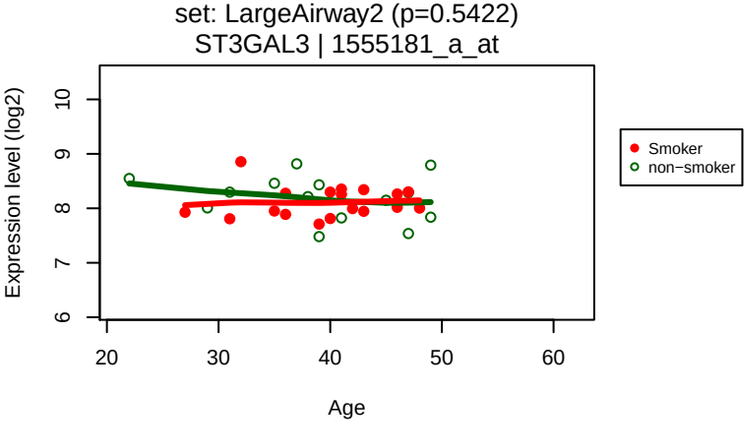

Gene expression ratio (smokers/nonsmokers)
gene: ST3GAL3 | 1555181_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1289
p=0.3437
p=0.1876
p=0.5584
p=0.5422
p=0.2896
p=0.094
p=0.5347
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

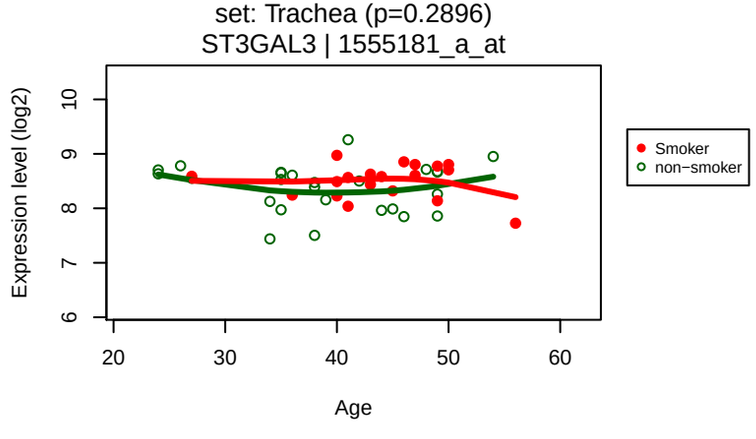

Gene expression ratio in Males
gene: ST3GAL3 | 1555181_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2604
p=0.0292
p=0.1061
p=0.4992
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST3GAL3 | 1555181_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.88
p=0.2454
p=0.9837
p=0.8922
SmallAirway2
LargeAirway2
Trachea
Macrophage2

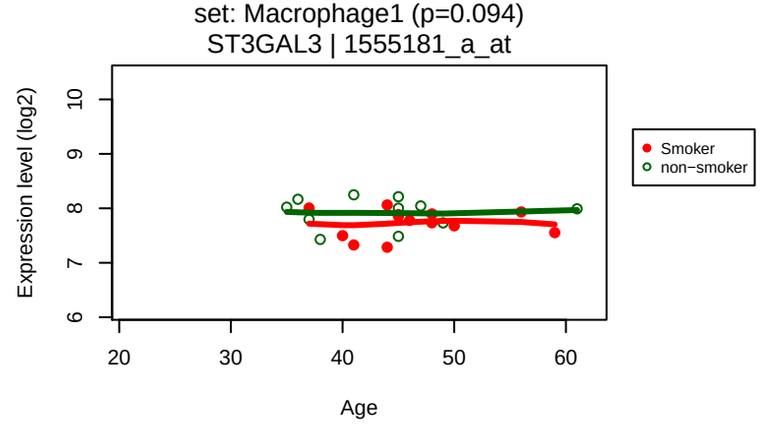

set: SmallAirway1 (p=0.1289)
ST3GAL3 | 1555181_a_at
Expression level (log2)
Age
Smoker
non−smoker

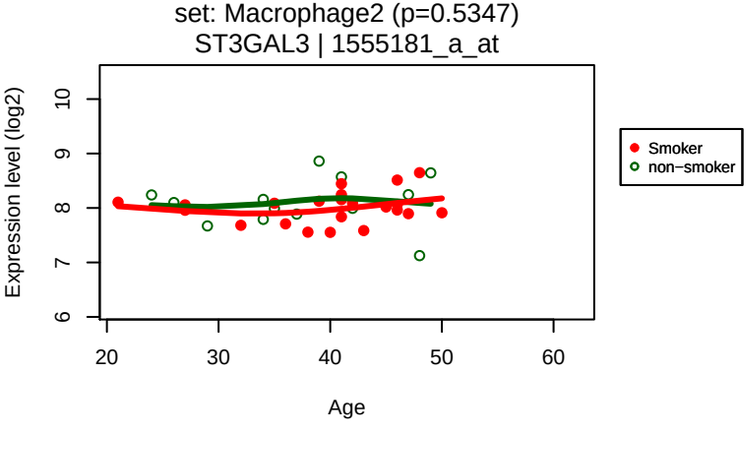

set: SmallAirway2 (p=0.3437)
ST3GAL3 | 1555181_a_at
Expression level (log2)
Age
Smoker
non−smoker

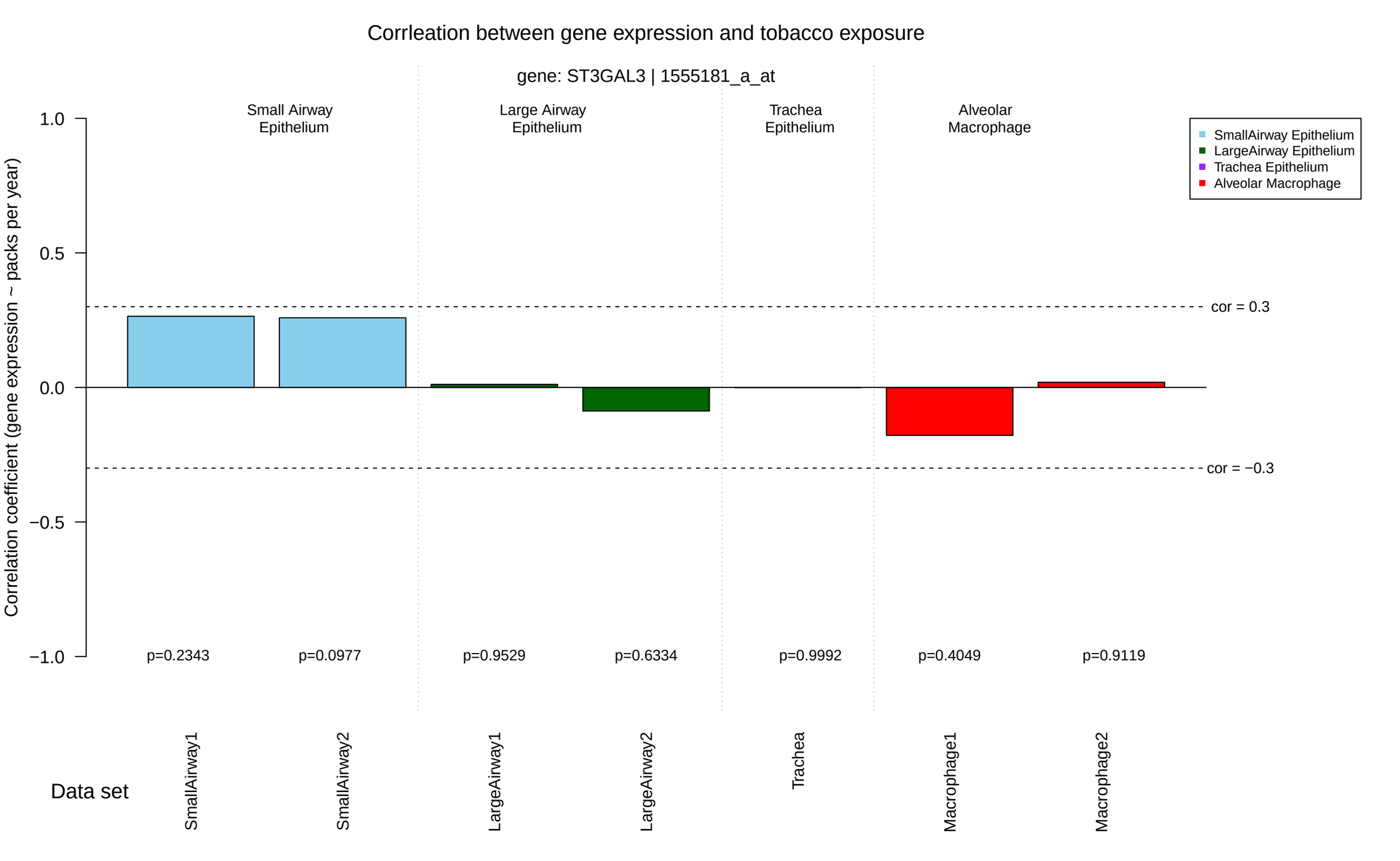

set: LargeAirway1 (p=0.5584)
ST3GAL3 | 1555181_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5422)
ST3GAL3 | 1555181_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2896)
ST3GAL3 | 1555181_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.094)
ST3GAL3 | 1555181_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5347)
ST3GAL3 | 1555181_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST3GAL3 | 1555181_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2343
p=0.0977
p=0.9529
p=0.6334
p=0.9992
p=0.4049
p=0.9119
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2